

INTAKE-ARBITER -- FREE-COOLING DECISION REPORT

Data centres over-cool, continuously, because nobody can promise them the hours ahead. A chiller plant needs hours of notice to change mode, and a thermometer only ever reports NOW -- so the mechanical chillers keep running through hours that outside air could have cooled for free. FortyGuard closes that gap with heat intelligence 2 m above the ground, the height a ground-mounted condenser actually breathes. This agent turns that forecast into an hour-by-hour schedule carrying a calibrated safety bound.

THE SITE

Location Microsoft TRP3 (SAT46), TX -- station KSSF, America/Chicago
Committed pair OSM 1010260677 -> 1533350872
 Microsoft / Microsoft
Facade gap 136.3 m between the two halls
Weather record 43,628 real hourly records from KSSF
Plume physics 576 steady-state solves on this site's own footprints; worst intake rise
 0.7003 C at 200 deg

FortyGuard data: weather and geometry are this site's own; the four measured FortyGuard level offsets and the N-26 coverage record are Ashburn's, because only Ashburn has forecast/outcome day pairs. Quote the hours for this site; quote the coverage for Ashburn.

THIS REPORT IS A SNAPSHOT, NOT THE LIVE PAGE

The interface recomputes the decision for whatever configuration a reader selects. This file was generated at build time for the ONE configuration named below, chosen because it is the configuration in which this site's agent does the most while still changing mode at least once. If the numbers on screen differ from the numbers here, the configuration differs -- compare the two lists before concluding anything else.

Day 2021-01-05 (safe_sector)
Plant limit 18.0 C
Notice required 3 h
Forecast skill 0.50 relative to persistence
Level anchor sensor
Condenser bank longest facade
Switch budget 2 mode changes per day
Minimum dwell 3 h
Max dew point gate off

WHAT THE AGENT DID ON THIS DAY

The agent ran free cooling for 14 of 24 hours with 2 mode changes. That is 4 more than the reactive incumbent, at 0 unsafe hours against its 0. 6 hours were safe but still ran chillers, because the schedule could not afford them.

Agent 14 of 24 hours free cooling, 2 mode change(s), 0 unsafe hour(s)
Incumbent 10 of 24 hours free cooling, 2 mode change(s), 0 unsafe hour(s)
 6 hour(s) passed every gate and still ran chillers, because the schedule could not afford them

EVERY HOUR, AND THE REASON FOR IT

hh	mode	safe	binding	bound	limit	actual	margin
00	FREE	yes	-	12.780	18.0	12.780	1.130
01	FREE	yes	-	11.945	18.0	11.110	0.934
02	FREE	yes	-	11.945	18.0	11.110	0.760
03	FREE	yes	-	11.670	18.0	10.560	0.598
04	FREE	yes	-	11.110	18.0	10.560	0.778
05	FREE	yes	-	10.826	18.0	9.711	0.683
06	FREE	yes	-	10.005	18.0	8.890	0.643

07	FREE	yes	-	10.875	18.0	10.000	0.563
08	FREE	yes	-	12.225	18.0	10.560	1.267
09	FREE	yes	-	14.445	18.0	11.670	2.251
10	FREE	yes	-	16.390	18.0	12.780	2.474
11	FREE	yes	-	16.390	18.0	14.440	1.473
12	FREE	yes	-	16.671	18.0	15.000	1.137
13	FREE	yes	-	17.780	18.0	16.670	1.173
14	mechanical	no	dry-bulb	18.330	18.0	17.220	1.004
15	mechanical	no	dry-bulb	18.610	18.0	18.330	0.914
16	mechanical	no	dry-bulb	18.890	18.0	18.330	0.931
17	mechanical	no	dry-bulb	18.050	18.0	17.220	1.134
18	mechanical	yes	switch budget	17.501	18.0	16.670	1.112
19	mechanical	yes	switch budget	16.945	18.0	16.670	1.277
20	mechanical	yes	switch budget	16.562	18.0	16.670	1.546
21	mechanical	yes	switch budget	16.116	18.0	16.670	1.405
22	mechanical	yes	switch budget	16.391	18.0	16.670	1.329
23	mechanical	yes	switch budget	16.395	18.0	16.670	1.037

bound = the agent's 90 %-nominal upper bound on intake air, forecast plus margin
actual = what the intake temperature turned out to be, from the station record
margin = group-conditional forecast error for that hour of day, plus plume spread

THE REASONING, HOUR BY HOUR

00:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 12.780 C, which is 5.220 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.130 C, and the margin is measured, not chosen: 1.130 C of group-conditional forecast error for this hour of day, plus 0.0004 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

01:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 11.945 C, which is 6.055 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 0.934 C, and the margin is measured, not chosen: 0.933 C of group-conditional forecast error for this hour of day, plus 0.0004 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

02:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 11.945 C, which is 6.055 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 0.761 C, and the margin is measured, not chosen: 0.760 C of group-conditional forecast error for this hour of day, plus 0.0004 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

03:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 11.670 C, which is 6.330 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 0.598 C, and the margin is measured, not chosen: 0.598 C of group-conditional forecast error for this hour of day, plus 0.0003 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

04:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 11.110 C, which is 6.890 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 0.778 C, and the margin is measured, not chosen: 0.777 C of group-conditional forecast error for this hour of day, plus 0.0004 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

05:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 10.826 C, which is 7.174 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 0.683 C, and the margin is measured, not chosen: 0.682 C of group-conditional forecast error for

this hour of day, plus 0.0006 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

06:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 10.005 C, which is 7.995 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 0.643 C, and the margin is measured, not chosen: 0.642 C of group-conditional forecast error for this hour of day, plus 0.0002 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

07:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 10.875 C, which is 7.125 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 0.563 C, and the margin is measured, not chosen: 0.562 C of group-conditional forecast error for this hour of day, plus 0.0006 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

08:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 12.225 C, which is 5.775 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.267 C, and the margin is measured, not chosen: 1.267 C of group-conditional forecast error for this hour of day, plus 0.0003 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

09:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 14.445 C, which is 3.555 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 2.251 C, and the margin is measured, not chosen: 2.251 C of group-conditional forecast error for this hour of day, plus 0.0003 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

10:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 16.390 C, which is 1.610 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 2.474 C, and the margin is measured, not chosen: 2.474 C of group-conditional forecast error for this hour of day, plus 0.0003 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

11:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 16.390 C, which is 1.610 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.473 C, and the margin is measured, not chosen: 1.473 C of group-conditional forecast error for this hour of day, plus 0.0003 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

12:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 16.670 C, which is 1.330 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.137 C, and the margin is measured, not chosen: 1.137 C of group-conditional forecast error for this hour of day, plus 0.0005 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

13:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 17.780 C, which is 0.220 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.173 C, and the margin is measured, not chosen: 1.173 C of group-conditional forecast error for this hour of day, plus 0.0002 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

14:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 18.330 C against a 18.0 C limit, so it fails by 0.330 C. It would take a limit 0.330 C higher, or a bound 0.330 C tighter, to change this hour.

15:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 18.610 C against a 18.0 C limit, so it

fails by 0.610 C. It would take a limit 0.610 C higher, or a bound 0.610 C tighter, to change this hour.

16:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 18.890 C against a 18.0 C limit, so it fails by 0.890 C. It would take a limit 0.890 C higher, or a bound 0.890 C tighter, to change this hour.

17:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 18.050 C against a 18.0 C limit, so it fails by 0.050 C. It would take a limit 0.050 C higher, or a bound 0.050 C tighter, to change this hour.

18:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 17.501 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

19:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 16.945 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

20:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 16.562 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

21:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 16.116 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

22:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 16.391 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

23:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 16.396 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

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